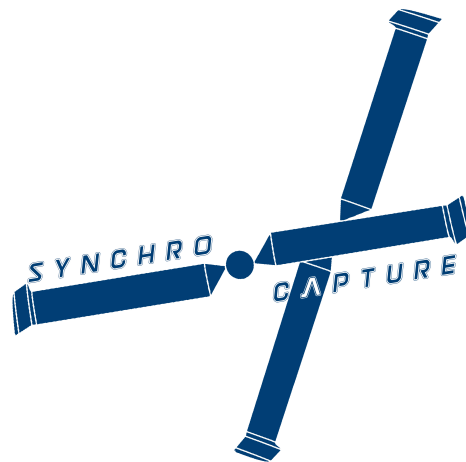


AERO96005 - GROUP DESIGN PROJECT
WHOLLY REUSABLE LAUNCH SYSTEM (RECOVERY)

DESIGN OF A NOVEL SYNCHROPTER CONCEPT FOR THE AERIAL-RECOVERY OF FALLING ROCKET STAGES



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Abstract

This third year Group Design Project (GDP) aims to develop a preliminary design for the mid-air recovery aspect of a wholly reusable launch system for which as much as possible can be recovered and reused for subsequent launches of LEO satellites. This report forms a subset of the marking criteria for the *AERO 96005 - Group Design Project* module, seeking to test engineering judgement and design trade-offs within group settings.

Covered in this document is the initial sizing of the recovery vehicle, the choice of structural layout and the design, analysis and material selection of the vehicle's rotor blades. The rotor blades designed have an estimated weight of 60kg and are predicted to not show resonant behaviour when in operation.

Acknowledgements

I must thank a handful of people for their help whilst making this report. Thank you to Lewis who was able to provide me with insight and advice, thank you to Atri who helped me solve the problem of why my helicopter blades were 10 times bigger than they needed to be and thank you to my team supervisor Zahra who was able to provide advice and guidance throughout the project whenever necessary.

WRR Team Members

Table 1: *Team Members and Roles.*

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Part I

Introduction

In recent years there has been an increase in activity in the commercial space sector with the number of rocket launches increasing compared to the past decade [6]. With this has come increased interest in reducing the overall cost of launching rockets, in particular from the reuse of components. While reusable rocket systems have already been developed there has been little development into the aerial recapture of said rockets. The advantage of such a proposal is that it allows for weight to be saved on the rocket, as no provisions need to be made for the rocket to land itself, meaning a larger payload can be carried into orbit. At the moment, current aerial recovery solutions have used existing vehicles that have been modified to suit individual requirements. This project instead attempts to design a purpose-built aerial vehicle to better meet these requirements and be more cost-effective than existing vehicles.

For the Structures and CAD sub-team the objective of this project was to design, analyse and model the primary structure of an aerial recovery vehicle intended to capture a payload that was descending from low Earth orbit (LEO). Outlined in this report is the conceptual design of the vehicle's structure and the preliminary design of the vehicle's rotor blades.

Part II

Pre-Conceptual Design

1 Choice of Recovery Vehicle

In the initial stages of the project 3 vehicle concepts were examined, they were a fixed wing aircraft, a helicopter and a tilt-rotor. In this report the weight estimate of the helicopter is examined.

2 Use of Regression Data

Data from over 20 helicopters [7] was analysed and a relationship between the installed power and maximum take-off weight was found. This relationship is illustrated in figure 1 with a line of best fit.

Installed power was chosen as a variable due to the mission requirement of the helicopter to be able to travel to an altitude of 20,000ft [8]. This meant it would require more power than traditional helicopters which would greatly impact the helicopter's proposed maximum take-off weight

For an initial power estimate of 1,118W for a 1 tonne payload an MTOW of 4,689kg was predicted [9]. However, it was noted that the data used for this estimate was made up of entirely manned aircraft and so a correction factor was applied to account for loss of weight due to the lack of passengers and ancillaries giving a new MTOW estimate of 2,500kg [8].

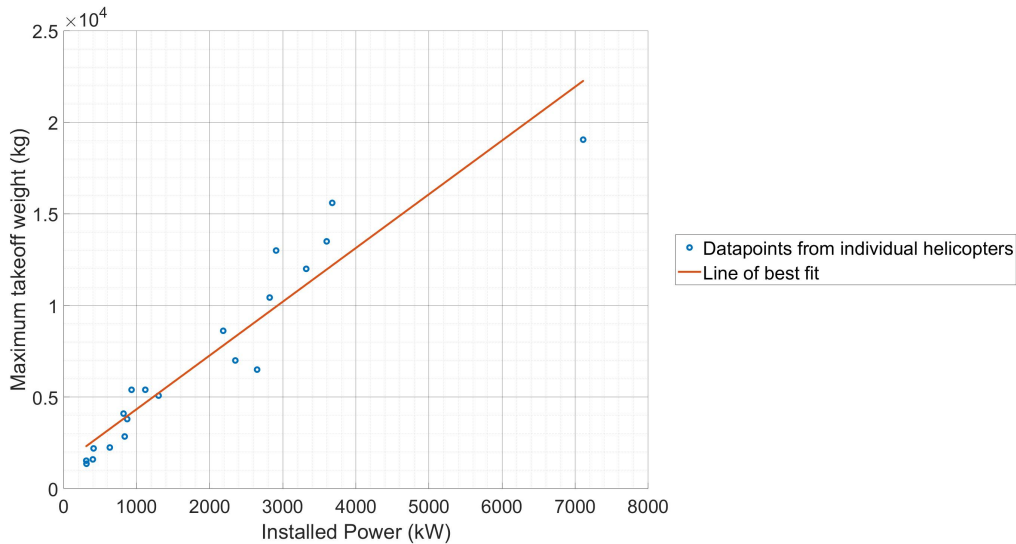


Figure 1: Plot of existing helicopters' installed power and MTOW with a line of best fit

Part III

Conceptual Design

3 Structural Design/Layout

Once an unmanned syncrocopter had been chosen as the recovery vehicle the overall structural layout was chosen. Firstly, existing helicopter designs were looked at and their advantages/disadvantages were noted. The designs considered were:

- Truss/tubular frame construction
- Monocoque/semi-monocoque construction
- Composite/bonded construction

The use of a tubular frame/truss design was considered as the method is well established as it has been used for construction of early helicopters and provides a strong, easy to analyse structure. The disadvantage of this method is its weight as it is heaviest of the all the designs considered and it is also hard to form into an aerodynamic shape due to its use rigid bars in making up the structure.

The next construction method that was analysed was the use of a monocoque/semi-monocoque design. This approach is also well documented and has the benefit of being lighter than a truss structure. However, while it is well suited for a manned vehicle it is not for an unmanned vehicle due to the large open space left inside the fuselage and so is spatially inefficient.

Finally, the use of composites was considered. More recently composites have been used extensively in helicopters from the rotor blades to the fuselage. They have the benefit of being lighter than metals and have a higher specific strength. They also exhibit good fatigue resistance which is especially important in the cyclic loading environment of a helicopter and they have good corrosion resistance which is important in a maritime environment which the helicopter is meant to operate in.

As the helicopter was to be unmanned it was decided to design the main body of the helicopter using a truss structure. The reason for this was to efficiently transfer the loading of all the different components in the fuselage while being spatially efficient. The trusses themselves were chosen to be made out of metal due to the ease of manufacture and joining of members together through welding [10]. An example of this structure being used in an existing unmanned helicopter can be seen in figure 2



Figure 2: Internal structure of an existing unmanned helicopter - ILX-27 [2]

To simplify the analysis of the fuselage skin it was decided to make an aerodynamic fairing which would fit around the truss structure and carry no significant loads. This allowed the Aerodynamics team to make a fuselage skin which suited their requirements whilst minimising their overall impact on the fuselage design.

Part IV

Preliminary Design

4 Blade Structural Design

4.1 Blade Modelling

Once the airfoil sections and chord lengths had been chosen for the main rotor by the Aerodynamics team [11] it was then modelled as a cantilever beam with a built-in support and free end due to the slenderness of the blade. The geometry of blade was carefully modelled as at the tip there was a swept section which when producing lift would generate a torsional load, so it was important to establish the magnitude of torque produced so that could be factored into the analysis.

The most constraining aerodynamic forces were also provided by the Aerodynamics team and were scaled by a load factor of -1.0 and 3.5 to ensure they would meet the loading requirements outlined in FAR-29 [12]. These forces were used to find the shear force and bending moment distribution along the blade. These distributions were checked to ensure they were correct. For instance, the shear force at the root of the blade was checked to ensure it was equal to the lift and drag. From there the bending moment was found from the integral of the shear force and both parameters were checked to ensure that they tapered off to zero at the free end of the blade. This allowed the stress

in the cross-section to be found using equation 1

$$\sigma_z = \frac{\bar{M}_x}{I_{xx}}\bar{y} + \frac{\bar{M}_y}{I_{yy}}\bar{x} \quad (1)$$

Where

$$\bar{M}_x = \frac{\left(M_x - \frac{I_{xy}}{I_{yy}}M_y\right)}{\left(1 - \frac{I_{xy}}{I_{yy}}\frac{I_{xy}}{I_{xx}}\right)} \quad \bar{M}_y = \frac{\left(M_y - \frac{I_{xy}}{I_{xx}}M_x\right)}{\left(1 - \frac{I_{xy}}{I_{yy}}\frac{I_{xy}}{I_{xx}}\right)} \quad (2)$$

Where

- M_x - x axis bending moment
- M_y - y axis bending moment
- I_{xx}, I_{yy}, I_{xy} - Second moment of area
- $\bar{x}, \bar{y}$ - Distance from the centroid in the x and y direction respectively

The effect of centrifugal force on the blade was also accounted for and the shear flow inside the blade was also calculated to ensure that both forces wouldn't cause material failure.

The vertical displacement of the blade was found by integrating equation 3 and these values were given to different teams to ensure sufficient clearance with different parts of the helicopter when the blades were at their maximum deflection. Deflections due to shear were ignored because due to the length of the beam shear deflections would only account for a small percentage of the overall deflection [13].

$$M = -EI \frac{d^2w}{dx^2} \quad (3)$$

Where

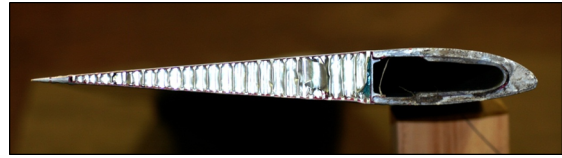
- M = x axis bending moment
- E = Blade material Young's modulus
- I = Second moment of area of blade cross-section
- w = Blade displacement

4.2 Blade Material Selection

The material selection of the blade was driven by two competing factors. These were low weight and high strength. When looking at historical rotor blades they were initially made of light-weight and strong metal alloys such as titanium or aluminium [14]. However, with the increased manufacturability of composites such as CFRP there was a shift towards making rotor blades out of a composite shell with a lightweight filling for rigidity such as honeycomb or foam [15]. One of the secondary considerations for the material selection was corrosion resistance due to the maritime operating environment of the helicopter. This meant it would be exposed to salt water which is more corrosive than fresh water [16]. Another consideration was the fatigue life of the material as helicopter rotors can produce vibrations [17] which may lead to crack propagation and subsequent material failure.



(a) Cross-section of a Lynx rotor blade made of composite materials [18]



(b) Cross-section of a Bell 206 rotor blade made of aluminium honeycomb [19]

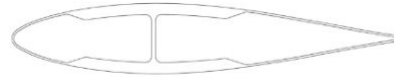
Figure 3: Cross-sectional views of two helicopter rotor blades showing internal materials and structure

Once the stresses inside the blade had been calculated the internal geometry was defined. A two cell construction with a dividing spar was chosen to increase the torsional rigidity of the blade. It was decided not to add any more cells due to the added complexity of manufacturing such a design and also the added redundancy of multiple cells wasn't necessary. This was because the helicopter was to be designed for civil applications whereas in military helicopters more cells may be required in order to resist damage.

To ensure efficient use of material throughout the blade section the decision was taken to utilise a design with thickened spar caps which would take most of the loading and therefore leave a reduced shell. Figure 4 shows an example of this in wind turbine blade cross sections where this design approach has been used to reduce the material used by 25% [3].



(a) Cross section of a typical wind turbine blade with uniform skin thickness



(b) Cross section of a wind turbine blade with reduced skin thickness and thicker spar caps

Figure 4: Wind turbine blade cross sections with the same flapwise stiffness but different weights. Section (b) uses 25% less material than section (a) [3]

4.3 Blade Cross Section

Once analysis of the blade was completed it was found to have stresses to allow the use of carbon fibre/epoxy composites in both the spar and the skin. To supplement the carbon fibre, foam was chosen as a filler material for the cavity inside the blade section to ensure that the blade would be able to maintain its shape under load and be more resistant to crushing loads. At the leading edge metal weights were also placed to ensure that the centre of mass was as close as possible to the shear centre to minimise any torque produced when the blade was loaded.

Additionally, it was chosen to wrap the blade in a fine copper mesh. This was because helicopter blades can suffer from lightning strikes [20] and without a conductive path to take away any current the blades can be damaged by the high temperatures and currents. At the leading edge a thermo-electric de-icing strip was added so that the rotor will be able to operate in cold weather at high altitudes and a stainless steel erosion strip was also added to ensure that the blade wouldn't be damaged by objects impacting the leading edge during operation. Combined this leads to a predicted single blade weight of 60kg. A proposed cross section of the rotor blade can be seen in figure 5

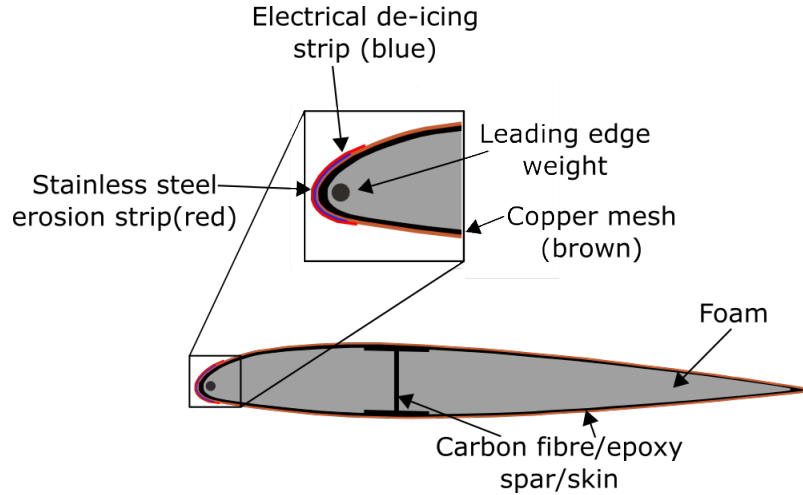


Figure 5: Cross section of the proposed rotor blade

4.3.1 Carbon fibre spar/skin

To achieve the desired material properties and thickness of the spar caps it was decided to make them out of 0° and 90° laminates. The material properties of the composite used are shown in table 2. The use of 0° laminates was done to ensure the structure would be able to withstand the predicted loads while some 90° laminates were used to prevent delamination and stop the propagation of cracks. The web was then chosen to be made out of 90° laminates i.e. fibres which run vertically up and down, due to the fact that the web will take shear force and see hardly bending. Finally these two components were joined together as shown in figure 6 through process of making the web and spar caps separately and then using C sections to ensure that the structure wouldn't come apart under loading. The skin of the section was made of a $[\pm 45^\circ]_S$ to provide some torsional resistance. Table 3 shows the proposed dimensions and layups for the composite spar.

Table 2: Material properties of the carbon fibre/epoxy composite used. Data taken from [1]

Material Property	Value
E_{11}	140 GPa
E_{22}	8.50 GPa
ν_{12}	0.330
ν_{21}	0.022
G_{12}	5.00 GPa
σ_{11y} - tension	2.10 GPa
σ_{11y} - compression	1.69 GPa

¹ 90° refers to the fibres having a vertical orientation so that they are in line with the direction of the web

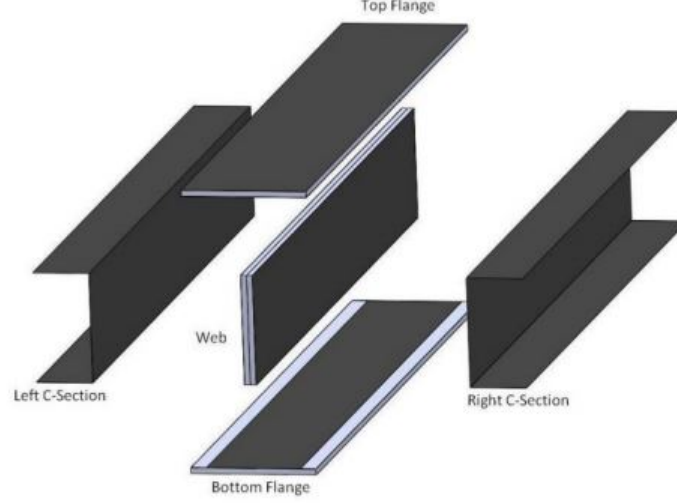


Figure 6: Manufacture of a composite I-beam showing how the spar caps and web are joined through the use of C sections [4]

Table 3: Table of key dimensions of carbon fibre/epoxy components in the helicopter blade

Property	Dimension
Spar web thickness	1.0 mm
Spar web layup ¹	$[90^\circ]_5$
Spar cap thickness	4.0 mm
Spar cap total width	150 mm
Spar cap layup	$[(0^\circ_2/90^\circ/0^\circ_2)_2]_s$

5 Blade Vibration

5.1 Vibration Modelling

When the rotor blades of a helicopter are in operation they can exhibit dynamic behaviour due to vibrations. Specifically, they can show lead/lag, flapping and feathering modes about each of their axes. To avoid resonance when operating the resonant frequencies of the blades were checked to ensure there would not be any resonant cross-over during the helicopter's operation at its target blade speed. To make the analysis simpler the modes considered were assumed to be decoupled even though this is not the case in real world operation. The three modes considered are shown in figure 7.

First the non-rotating natural frequency of the blade was found using equation 4 taken from [21]

$$\omega_{nf} = \alpha_n^2 \sqrt{\frac{EI}{\rho AL^4}} \quad (4)$$

Where

- α_n - Constant dependent on the mode being examined - for the first and second natural frequencies the value is 1.875 and 4.694 respectively [21]
- E - Blade material Young's modulus

- I - Second moment of area of blade cross-section
- A - Blade cross-sectional area
- ρ - Blade material density
- L - Length of the blade

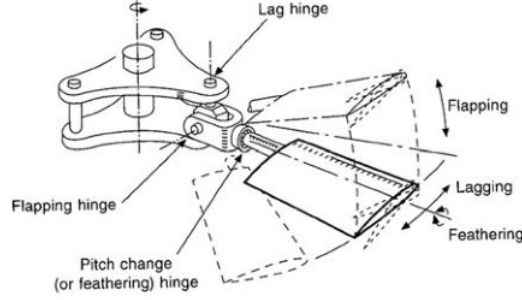


Figure 7: Hinges on a typical rotor showing the three different degrees of freedom [5]

5.2 Lead-lag Frequency

Using the natural frequency from equation 4 the lead-lag frequency of the blade at various speeds was calculated using equation 5 taken from [22]

$$\omega_{\text{lead-lag}} = \omega_{nf} + \phi\Omega^2 \quad (5)$$

Where ϕ is a function of the desired mode shape [22]. For the first mode ϕ was found using equation 6 and for the second using equation 7

$$\phi = 0.173 + 1.558 \frac{R}{L} \quad (6)$$

$$\phi = 5.38 + 8.631 \frac{R}{L} \quad (7)$$

Where

- R - Distance from centre of rotation to start of blade
- L - Length of blade
- Ω - Rotational velocity

5.3 Flapping Frequency

The flapping frequencies of the blade were found using equation 8 taken from [23] where the natural frequencies were calculated using equation 4

$$\omega_{\text{flapping}}^2 = 1 + \left(1.6102 \frac{R}{L} + 1.1964\right) \left(\frac{\Omega}{\omega_{nf}}\right)^2 \quad (8)$$

Where

- R - distance from centre of rotation to start of blade
- L - Length of blade
- Ω - Rotational velocity

5.4 Feathering Frequency

The torsional feathering frequencies were found in a similar way to the previous two modes. First the non-rotating torsional mode of the blade was found using equation 9 taken from [24]. This non-rotating frequency was then used to find the natural frequency during rotation using equation 10 taken from [24] as well.

$$\omega_0 = \left(m - \frac{1}{2}\right) \pi \sqrt{\frac{GJ}{IR^2}} \quad (9)$$

Where

- m - Mode number - 1,2,3...
- R - Radius of beam
- GJ - Torsional rigidity of the blade
- I - Torsional mass moment of inertia about the elastic axis

$$\omega_{\text{feathering}} = \sqrt{(\omega_0^2 + \Omega^2)} \quad (10)$$

Where

- ω_0 - Non-rotating natural torsional frequency calculated from equation 9
- Ω - Rotational velocity

5.5 Blade Resonant Modes

The behaviour of each of the blades resonant modes at various rotational velocities are shown in figure 8. The resonant frequencies of each mode normalised with respect to the rotor rotational velocity are shown in table 4

These results show that when the rotor is operating at its target speed none of the resonant frequencies of any of the modes are near any integer multiples of the blade's rotational velocity. As such it is not expected that the blade will display resonance when in operation.

Table 4: Table of the first two resonant frequencies of each resonant mode normalised with respect to the helicopter rotor's angular velocity

Mode	Resonant frequency at rotor angular velocity
1st Flap	1.28
1st Lag	0.79
1st Torsion	2.29
2nd Flap	3.78
2nd Lag	4.37
2nd Torsion	6.27

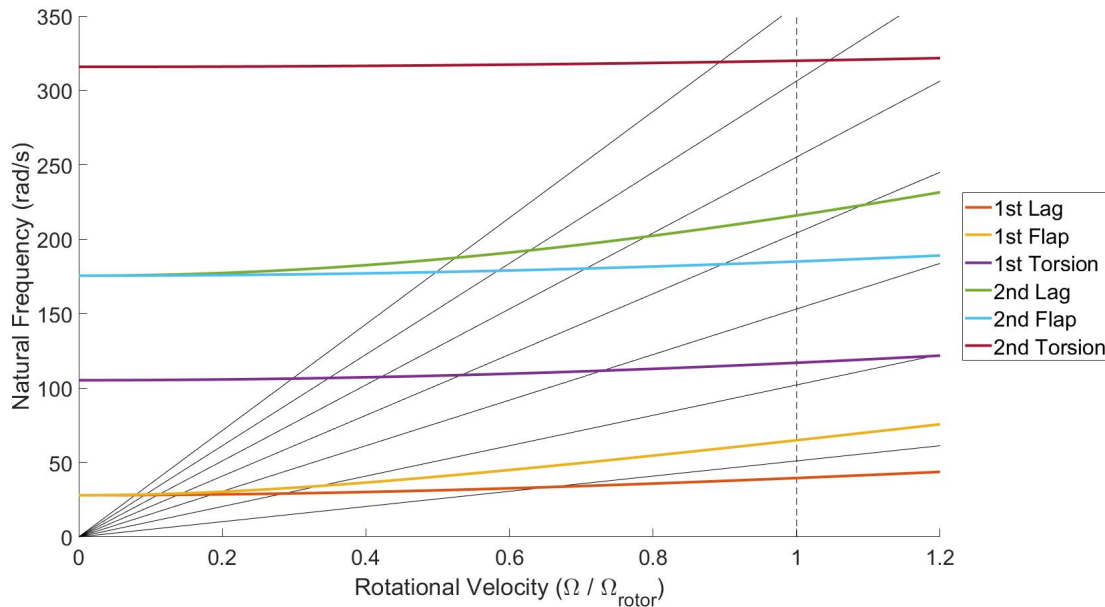


Figure 8: Fan plot showing the first two resonant frequencies of each mode for the helicopter blade. The black lines emanating from the origin are integer multiples of the blade's rotational velocity

Part V

Discussion of Design

6 Blade Structural Design

Although the structural analysis of the blade produced reasonable values of skin thickness and material stresses there were aspects that weren't in line with predictions from literature. One instance of this is the inclusion of centrifugal force into the analysis. According to [25] the centrifugal force in a helicopter blade should be the dominating load to the point where there shouldn't be any compressive stress in the blade. Yet, during the analysis carried out in this report the opposite was found to be true with the stresses due to centrifugal loading being orders of magnitude lower than the stress produced due to aerodynamic loading. Because of this, if the design were to be continued a more in-depth look into the loading of the blade would need to be carried out in order to assess if the calculated forces acting on the blade were accurate.

7 Blade Resonant Frequencies

The calculated resonant frequencies of the blade seem to be consistent with work done by other groups designing similar helicopters [26]. This is particularly promising as the work done in [26] uses an in-house code which has been designed specifically to work with helicopters whilst the work carried out in this report is nowhere near as specialised. Therefore it shows that the assumptions made in this report can be justified to some extent as the calculated frequencies of the blade are not unreasonable.

Part VI

Limitations of Design

Whilst the analysis carried out has been performed to ensure accurate results there are limitations of the methods used as well as assumptions that were made that should be noted if the design process were to be continued

8 Blade Structural Design

When analysing the rotor's structure some assumptions were made about the loading conditions and geometry. The loading along the beam was assumed to act through the shear centre and this meant the effect warping was not considered. Additionally, no dynamic loads were considered. This was neglected due to the complexity of analysing aeroelastic effects on the blade and other dynamic phenomena.

Another loading condition that was neglected was cyclic loading through vibration which determines the blades fatigue life. This was left out due to the difficulty in quantifying the magnitude of fatigue loading during the blade's predicted lifetime and subsequently its mean time to failure.

The connection of the blade onto the rotor hub hasn't been accounted for in this report as it assumed that the rotor was a rigid beam that ended at a location prescribed by the geometry given by the Aerodynamics team. If further work were to be done an interdisciplinary team could be setup to model the hub itself as well as the connection of the hub to the blade.

While the materials inside the blade's cross section have been designed to handle any compressive loads it is likely to see a buckling analysis hasn't been carried out on the blade's components. Specifically, the skin that wraps around the section and the spar caps. This is because of the difficulty of modelling such behaviour due to the unknown boundary conditions the components are likely to see. For instance, when operating normally some portion of the outer skin is in compression so there is a risk of buckling yet this skin is to be bonded to the foam underneath it. Therefore properties about the adhesive used would need to be known as well as how well the adhesive bonds to the carbon fibre as voids/imperfections would change the buckling strength of the skin. The same argument also applies for the spar cap. Because of these complexities this analysis has been discarded.

9 Blade Manufacturability

An area which has had little consideration in this report was the manufacturability of the rotor blade. While the rotor blade was made of materials which can be commercially produced the blade's geometry such as twist and the profile of the tip would make it hard to form the layers of carbon fibre into the desired shape.

10 Blade Vibration Modelling

When analysing the resonant modes of the rotor blades it was assumed that the modes were all uncoupled to make the analysis easier. However, this assumption is incorrect and because of the

coupling the resonant frequencies of a real rotor blade would be different. Another assumption made was that the blade was a homogeneous beam with uniform mass distribution and a constant second moment of area. Furthermore, in this simplified analysis some of the equations used were only approximations of a beam's real behaviour and in some cases they are only linear approximations.

Should further analysis be completed a more in-depth model of the coupling of the modes in the blade would be appropriate as well as the use of equations which better captured the blade's behaviour. Also, this model should account for change's in the blade's geometrical and physical properties as they would also influence its resonant modes.

Part VII

Conclusion

In this report the rotor blades for a syncrocopter have been designed to ensure they can withstand the aerodynamic loads expected during their operation. They are made out of lightweight materials in order to minimise their impact on the helicopter's weight and the use of carbon fibre ensures the blades have enhanced fatigue and corrosion resistance compared to metals. These blades have also been designed to ensure there is no resonance when operating meaning the helicopter's structure is unlikely to see high loading associated with such behaviour.

If more work were to be completed a more in-depth analysis of the forces that the blades are likely to see would be recommended as aeroelastic and other dynamic effects are not accounted. Furthermore, a finite element (FE) analysis could be carried out to analyse the stress distribution inside the blade and see how it compares to the calculations performed in this report. Additionally, although fatigue loading has been considered through the material selection of the blade, the magnitude of vibratory loading has not been quantified and so a more thorough analysis of the fatigue life would need to be carried out to see how suitable the current design is for implementation on a real helicopter. Finally, the coupled resonant modes of the blades would require more analysis to ensure no resonance would during operation.

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